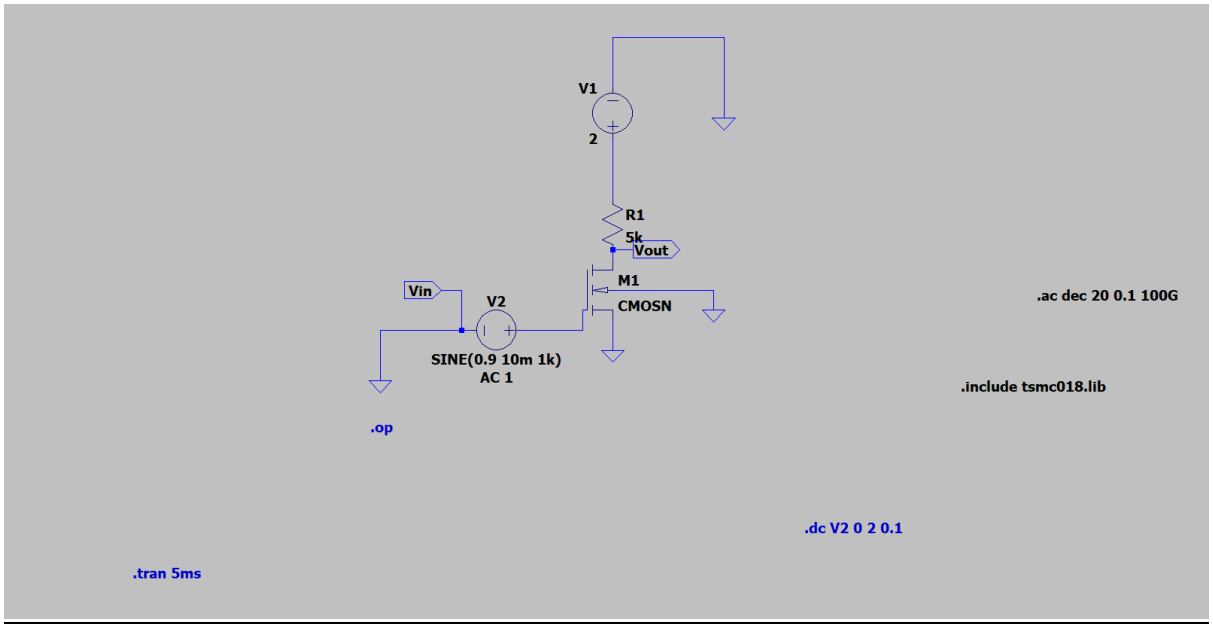
EXPERIMENT-1

Aim: To carry out DC, Transient, and AC analyses of a Common Source amplifier circuit in LTSpice and determine its parameters. Required Components: TSMC 180nm NMOS transistor (CMOSN), a drain resistor, and two voltage sources.

Circuit Diagram:



Parameters given are :

VDD= 2V

P<=1.2mw

Cl=0.7pf

Length=180nm

Vth=0.36

Procedure:

1. Construct the CS amplifier in LTSpice according to the reference circuit diagram.
2. Assign the given parameter values to their respective components.
3. Save the LTSpice simulation file, along with its associated library file, into a newly created directory.
4. Determine the required drain current given a power consumption of 1.2mW.

Calculations:

P = Vx I ----> (1)

From (1) we get,

I =P/V

I = 0.6 mA [assuming I is 200 mA coz p<= 1.2mv]

Vout = VDD – I x Rd ----> (2)

Vout = Vds (VDD / 2) -----> (3)

Now, substitute (3) in (2)

VDD / 2 = VDD – I x Rd

From the above equation we get Rd as 5kohm. Then we find W from Id equation.

Id = kn x 0.5 (Vgs – Vth) ^ 2 ----> (4)

The given values:- Vgs = 0.9 v,

Vth = 0.36 v,

Id = 200 mA,

Un = 273.8094484,

Tox = 4.110^-9,

Eox = 3.4530610^-11.

From the 4th equation we get W = 1.07072 um. Now enter this value in LTspice and execute the dc analysis.

<u>DC Analysis :</u>

We perform a DC analysis to determine the DC operating point and verify that the MOSFET is working in the saturation region.

1. Once the circuit is built, set the DC voltages to **VDD = 2V** and **Vgs = 0.9V**.
2. Navigate to the **Simulate** tab, select the **DC op pnt** (DC operating point) analysis, and click **OK**.
3. Click **Run** to execute the simulation.
4. Review the generated DC operating point values.

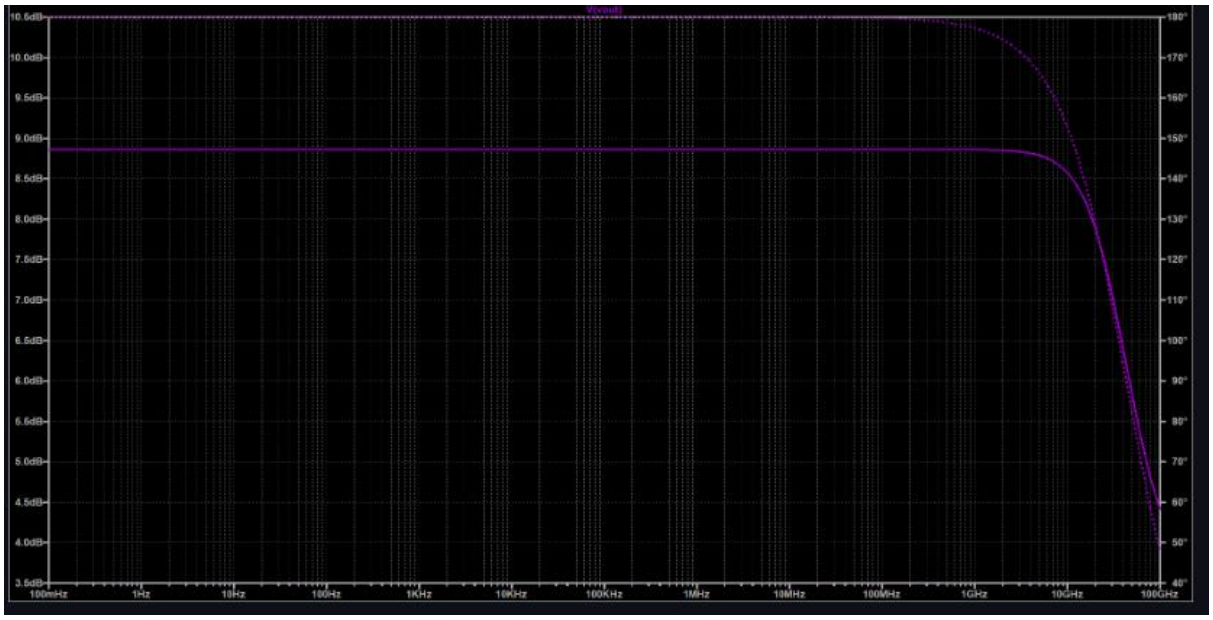
```
       --- Operating Point ---

V(n001):        2              voltage
V(n002):        0.9            voltage
V(vout):        1.06671        voltage
I(R1):          0.000186658    device_current
I(V1):          -0.000186658   device_current
I(V2):          0              device_current
Ib(M1):         -1.07671e-12   device_current
Id(M1):         0.000186658    device_current
Ig(M1):         0              device_current
Is(M1):         -0.000186658   device_current
```

From the LTspice simulation we get I as 186 mA, so now we have to vary the width to get I as 200 mA. If width increases, current also will increase. In other words they are directly proportionally each other.

At W = 1.337u we get I as 200mA.

<u>AC analysis:</u>

1. Navigate to the **Simulate** menu and select **AC Analysis**. Set the type of sweep to **Decade**, the number of points per decade to **20**, the start frequency to **0.1Hz**, and the stop frequency to **1THz**.
2. Click **OK** and click **Run** to generate the circuit's frequency response and gain plot.



Gain = -gm x Rd

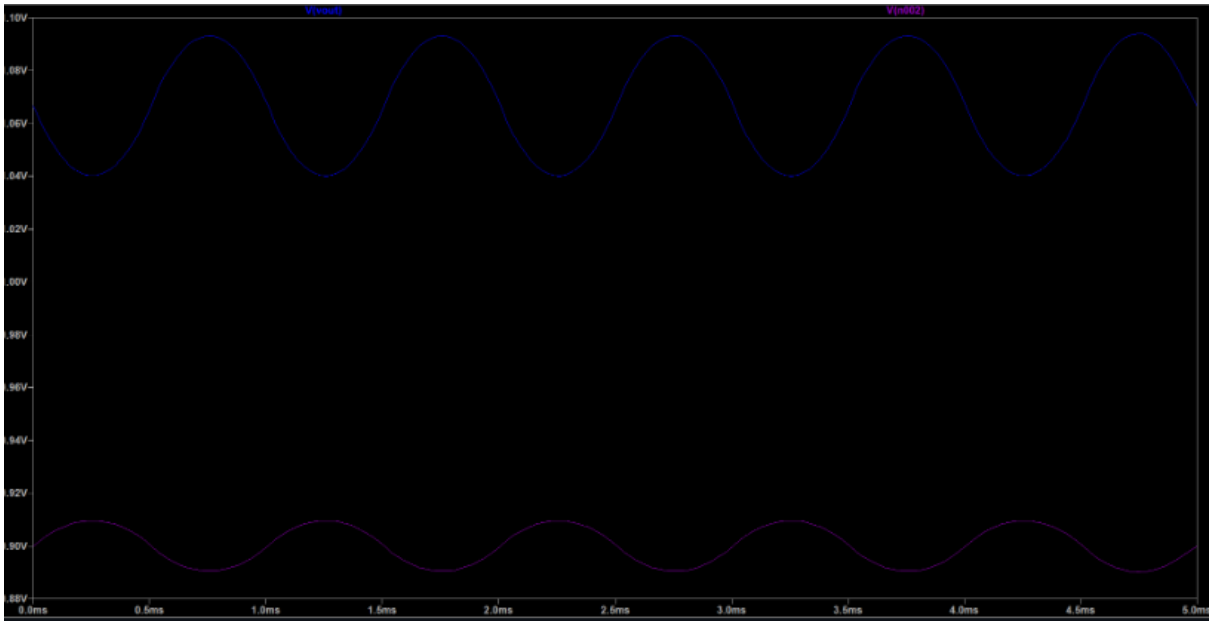
Gain = 2I x Rd / Vov

Gain = 3.7037 V

Gain in dB = 11.37201

<u>Transient Analysis:</u>

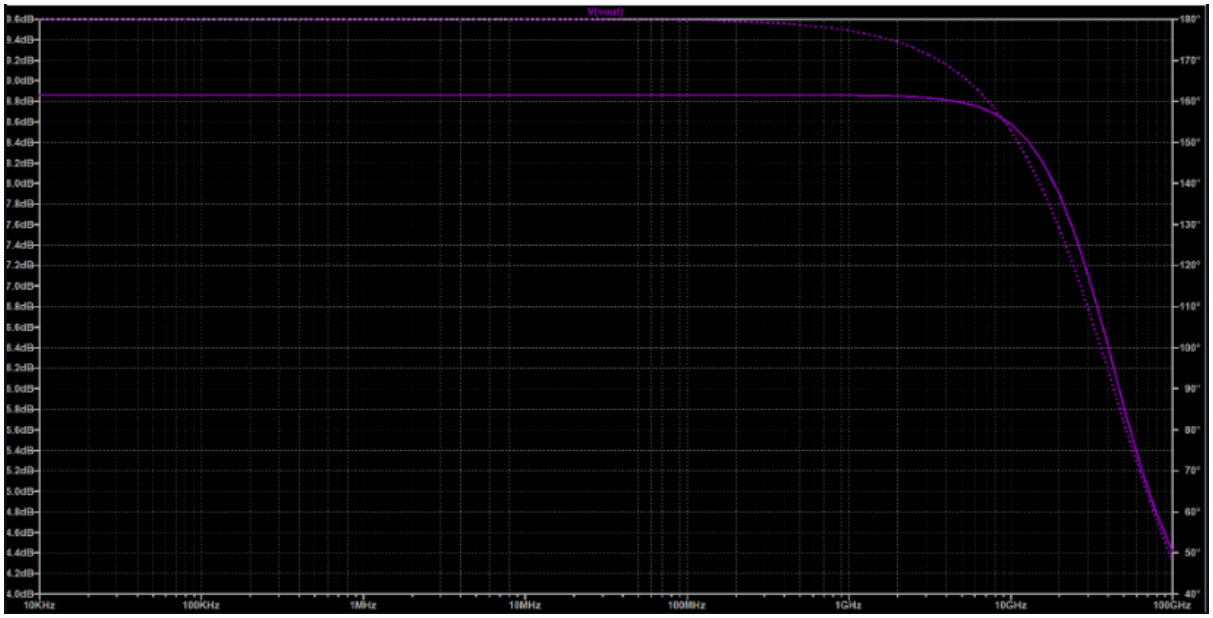
1. Configure the input source as a sine wave with a **1kHz** frequency, a **10mV** amplitude, and a DC offset (**Vgs**) of **0.9V**.
2. Navigate to the **Simulate** menu, choose **Transient Analysis**, and set the Stop Time to **5ms** (this creates the .tran 5ms directive).
3. Run the simulation to view how the circuit responds to the time-varying input signal.



Av = gain = Vout / Vin

Av = 2.7756 , Av in dB = 8.86713

Bandwidth = fh – fl = 22.2336 GHz.



Inference:

DC Analysis: The drain current is directly proportional to the width of the MOSFET, though variations can also occur due to channel length modulation. If your operating current is too low, you must increase the width (W) to achieve proper biasing.

Transient Analysis: This simulation is used to evaluate signal amplification, identify any DC shift between the input and output, and detect signal distortion. If the output is distorted, check the drain-to-source voltage (Vds) to ensure the MOSFET is still operating in the saturation region. If the output amplitude is too small, the circuit's gain needs to be increased.

AC Analysis: This is a small-signal analysis used to determine the overall gain of the circuit. We observe that there is low gain due to low value of gm and rd. Increase gm and Rd to increase gain of the circuit.

Conclusion:

A Common Source (CS) amplifier was successfully designed and simulated using LTSpice. Through DC analysis, the MOSFET was verified to be operating correctly in the saturation region. Transient analysis demonstrated the circuit's ability to amplify a time-varying signal, while AC analysis confirmed the overall voltage gain and frequency response. Ultimately, the simulations successfully validated the theoretical operation of the amplifier.